

The Full Asymptotic Transseries of the Fubini Numbers

Exact Pole Sectors, General Bell-Polynomial Coefficients, Nonlinear Transforms, and Rigorous Remainders

A self-contained research article with complete derivations

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Abstract

The Fubini numbers, also called ordered Bell numbers or preferential-arrangement numbers, are

$$F_n = \sum_{j=0}^n j! \left\{ \begin{matrix} n \\ j \end{matrix} \right\} \quad (n \geq 0),$$

and have exponential generating function $(2 - e^z)^{-1}$. Their familiar leading estimate $F_n \sim n!/[2(\log 2)^{n+1}]$ uses only the nearest singularity $z = \log 2$. The generating function actually has the complete vertical lattice of simple poles $\zeta_k = \log 2 + 2\pi i k$, $k \in \mathbb{Z}$. Summing their contributions gives an exact, convergent exponential transseries rather than merely a first-order singularity estimate.

Writing $\rho = \log 2$, $u_k = \rho/(\rho + 2\pi i k)$, and $\Gamma^*(n) = n!/[\sqrt{2\pi n}(n/e)^n]$, we prove the exact factorization

$$F_n = \frac{\sqrt{2\pi n}}{2\rho} \left(\frac{n}{e\rho} \right)^n \Gamma^*(n) \sum_{k \in \mathbb{Z}} u_k^{n+1} \quad (n \geq 1).$$

The pole sum is absolutely convergent and contains every exponentially small oscillatory sector. The common algebraic fluctuation factor has the all-orders Stirling expansion $\Gamma^*(n) \sim \sum_{j \geq 0} c_j n^{-j}$, where the coefficients are given explicitly by complete exponential Bell polynomials, integer partitions, and a triangular Bernoulli recurrence. Thus all mixed transseries coefficients are $C_{j,k} = c_j \rho/(\rho + 2\pi i k)$.

We supply explicit two-parameter remainder bounds for simultaneous truncation of the algebraic order and the pole lattice; an exact Borel–Laplace representation of the common Stirling factor; convergent multi-index formulae for logarithms, reciprocals, arbitrary analytic functions of the normalized sequence, and the ratio nF_{n-1}/F_n ; and a uniform extension to the Fubini polynomials $F_n(x) = \sum_j j! \left\{ \begin{matrix} n \\ j \end{matrix} \right\} x^j$, $x > 0$. A final meromorphic-EGF theorem explains why simple poles have constant sector amplitudes and how higher-order Fubini generating functions produce polynomial-in- n amplitudes governed by generalized Bernoulli numbers.

Master result

Put

$$\rho = \log 2, \quad \zeta_k = \rho + 2\pi i k, \quad \Lambda_k = \text{Log} \left(\frac{\zeta_k}{\rho} \right), \quad k \in \mathbb{Z},$$

with the principal logarithm. For every integer $n \geq 1$,

$$F_n = \frac{n!}{2} \sum_{k \in \mathbb{Z}} \zeta_k^{-n-1}$$

$$= \frac{n!}{2\rho^{n+1}} \left[1 + 2 \sum_{k \geq 1} \left(\frac{\rho}{|\zeta_k|} \right)^{n+1} \cos((n+1) \arg \zeta_k) \right].$$

This is an *exact convergent pole-lattice transseries*. If

$$\exp \left(\sum_{r \geq 1} \frac{B_{r+1}}{r(r+1)} t^r \right) = \sum_{j \geq 0} c_j t^j,$$

where B_m are the Bernoulli numbers and the vanishing odd Bernoulli numbers are understood, then the elementary-scale transseries is

$$F_n \sim \frac{\sqrt{2\pi n}}{2\rho} \left(\frac{n}{e\rho} \right)^n \sum_{k \in \mathbb{Z}} e^{-n\Lambda_k} \sum_{j \geq 0} \frac{C_{j,k}}{n^j}, \quad C_{j,k} = c_j e^{-\Lambda_k} = c_j \frac{\rho}{\zeta_k}.$$

The k -sum is convergent for every $n \geq 1$; the j -sum is the usual Poincaré expansion of the scaled gamma factor and is interpreted asymptotically or by its positive-axis Borel sum.

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Notation and conventions

Throughout,

$$\rho := \log 2, \quad \zeta_k := \rho + 2\pi i k \quad (k \in \mathbb{Z}).$$

The principal logarithm is denoted by Log . Bernoulli numbers use

$$\frac{t}{e^t - 1} = \sum_{m \geq 0} B_m \frac{t^m}{m!}, \quad B_1 = -\frac{1}{2}.$$

The complete exponential Bell polynomials Y_j are normalized by

$$\exp\left(\sum_{r \geq 1} x_r \frac{t^r}{r!}\right) = \sum_{j \geq 0} Y_j(x_1, \dots, x_j) \frac{t^j}{j!}.$$

A sum over $k \in \mathbb{Z}$ is ordinary and absolutely convergent whenever the summand is $O(|k|^{-1-\varepsilon})$. At exponent one we write $\sum_{k \in \mathbb{Z}}^{\text{sym}}$ for the symmetric limit $\lim_{K \rightarrow \infty} \sum_{k=-K}^K$.

The symbol $\sim$ has two related uses. Between functions it denotes asymptotic equivalence. Between a function and a formal inverse-power series it denotes a Poincaré asymptotic expansion. All exact identities are marked with equality, in particular the complete pole-lattice sum.

1 Introduction and scope

1.1 From one dominant pole to a complete transseries

The elementary derivation of the leading asymptotic formula for F_n is a standard application of singularity analysis. The nearest singularity of $(2 - e^z)^{-1}$ to the origin is the positive real pole $\rho = \log 2$, whose residue is $-1/2$. Therefore

$$F_n \sim \frac{n!}{2\rho^{n+1}}.$$

This statement is correct but discards far more structure than it retains. Every solution of $e^z = 2$ is a pole, and those solutions form the lattice

$$\rho + 2\pi i\mathbb{Z}.$$

The nonreal poles occur in conjugate pairs. Their coefficient contributions are exponentially smaller than the real-pole contribution and oscillate with n . If $q_k := \rho/|\rho + 2\pi i k|$, the principal-pole approximation has error $O(q_1^{n+1})$, whereas retaining the first conjugate pair leaves $O(q_2^{n+1})$; the improvement factor is therefore $O((q_2/q_1)^{n+1})$. Retaining the entire lattice gives an exact formula.

The word *full* in the title is used in the following precise sense.

1. Every singularity of the Fubini exponential generating function is represented by a transseries sector.
2. Every algebraic inverse-power coefficient multiplying every sector is given in closed form and by an effective recurrence.
3. Fixed finite truncations in both directions come with rigorous error estimates.
4. The factorial fluctuation may either be retained exactly, expanded to arbitrary algebraic order, or reconstructed by an exact Borel–Laplace integral on the positive axis.

Thus no pole sector is omitted. The only divergent object is the familiar Stirling series that appears when the exact factor $n!$ is replaced by an elementary expression. Keeping $n!$ or $\Gamma^*(n)$ exact removes even that divergence.

1.2 Three normalizations that should not be conflated

It is useful to separate three layers:

$$\begin{aligned} \text{exact factorial normalization: } & \frac{2\rho^{n+1}}{n!} F_n, \\ \text{elementary normalization: } & \frac{2\rho}{\sqrt{2\pi n}} \left(\frac{e\rho}{n}\right)^n F_n, \\ \text{logarithmic normalization: } & \log F_n - \left((n + \tfrac{1}{2}) \log n - n - (n + 1) \log \rho + \tfrac{1}{2} \log(2\pi) - \log 2\right). \end{aligned}$$

In the first normalization the answer is an exact convergent sum of exponentials with constant amplitudes. In the second, every exponential sector is multiplied by the same Stirling series. In the third, the algebraic and exponential corrections add, but products of pole sectors appear because of the logarithm. A substantial part of the paper is devoted to making these translations explicit.

Normalization point

A simple pole contributes *one exact constant-amplitude sector*. It does not produce an independent infinite $1/n$ fluctuation series. The inverse powers in the elementary normalization come solely from replacing $n!$ by Stirling's expansion. Higher-order poles do produce polynomial-in- n sector amplitudes; this distinction is proved in Section 9.

1.3 Relation to coefficient calculus

The derivation uses a small number of reusable coefficient-calculus principles:

1. labelled set partitions turn the Stirling sum into a geometric series of exponential generating functions;
2. the Mittag-Leffler decomposition turns a meromorphic generating function into a sum of elementary coefficient contributions;
3. exponentiating the Bernoulli expansion of $\log \Gamma^*$ turns logarithmic coefficients into complete Bell polynomials;
4. applying an analytic function to a convergent exponential correction turns ordinary powers into multinomial or multi-index coefficients.

These are exactly the points at which Stirling numbers, Bell polynomials, formal composition, and transseries algebra meet.

2 Combinatorial and probabilistic foundations

2.1 Ordered set partitions

Definition 2.1. An *ordered set partition* of an n -element labelled set is a sequence $(B_1, \dots, B_j)$ of nonempty pairwise disjoint blocks whose union is the whole set. Equivalently, it is a weak ordering: elements in the same block are tied, while the blocks themselves are linearly ordered.

Theorem 2.2 (Stirling transform). *The number of ordered set partitions of an n -element labelled set is*

$$F_n = \sum_{j=0}^n j! \left\{ \begin{matrix} n \\ j \end{matrix} \right\}.$$

In particular,

$$F_0 = 1, \quad F_1 = 1, \quad F_2 = 3, \quad F_3 = 13, \quad F_4 = 75,$$

and the sequence continues 541, 4683, 47293, 545835, ...

Proof. First partition the labelled set into j nonempty unordered blocks. There are $\left\{ \begin{smallmatrix} n \\ j \end{smallmatrix} \right\}$ choices. Then put the j blocks in one of their $j!$ linear orders. Summation over every possible number of blocks gives the formula. The empty set has one empty ordered partition, accounting for $F_0 = 1$. $\square$

2.2 Exponential generating function

Theorem 2.3 (Fubini EGF). *The exponential generating function is*

$$\mathcal{F}(z) := \sum_{n \geq 0} F_n \frac{z^n}{n!} = \frac{1}{2 - e^z}.$$

Proof. For fixed j , the standard exponential generating function of partitions into j nonempty blocks is

$$\sum_{n \geq 0} \left\{ \begin{smallmatrix} n \\ j \end{smallmatrix} \right\} \frac{z^n}{n!} = \frac{(e^z - 1)^j}{j!}.$$

Multiplication by $j!$ orders the blocks. Therefore, as a formal power series,

$$\mathcal{F}(z) = \sum_{j \geq 0} (e^z - 1)^j = \frac{1}{1 - (e^z - 1)} = \frac{1}{2 - e^z}.$$

The same calculation is analytic whenever $|e^z - 1| < 1$, and analytic continuation gives the displayed meromorphic function. $\square$

Corollary 2.4 (Binomial recurrence). *For $n \geq 1$,*

$$F_n = \sum_{j=0}^{n-1} \binom{n}{j} F_j = \sum_{j=1}^n \binom{n}{j} F_{n-j}.$$

Proof. The identity $(2 - e^z)\mathcal{F}(z) = 1$ gives, in positive degree,

$$2F_n - \sum_{j=0}^n \binom{n}{j} F_j = 0.$$

The $j = n$ summand is F_n ; moving it to the other side proves the first form. Replacing j by $n - j$ gives the second. $\square$

2.3 A Dobinski-type moment formula

The geometric expansion of the EGF already anticipates the pole formula:

$$\frac{1}{2 - e^z} = \frac{1}{2} \frac{1}{1 - e^z/2} = \frac{1}{2} \sum_{m \geq 0} \frac{e^{mz}}{2^m}.$$

Coefficient extraction gives the following exact analogue of Dobinski's formula for ordinary Bell numbers.

Theorem 2.5 (Geometric moment representation). *For every $n \geq 0$,*

$$F_n = \frac{1}{2} \sum_{m=0}^{\infty} \frac{m^n}{2^m},$$

where $0^0 = 1$. For $n \geq 1$ this is equivalently

$$F_n = \frac{1}{2} \operatorname{Li}_{-n} \left(\frac{1}{2} \right).$$

Proof. Expand each exponential in the preceding geometric series and interchange the absolutely convergent sums in a neighborhood of $z = 0$:

$$\frac{1}{2} \sum_{m \geq 0} 2^{-m} e^{mz} = \sum_{n \geq 0} \left(\frac{1}{2} \sum_{m \geq 0} \frac{m^n}{2^m} \right) \frac{z^n}{n!}.$$

Comparison with Theorem 2.3 proves the result. When $n \geq 1$ the $m = 0$ term vanishes, and the remaining series is the defining series for the polylogarithm at negative integer order. $\square$

Interpretation

Let G be the geometric random variable

$$\Pr(G = m) = 2^{-m-1}, \quad m = 0, 1, 2, \dots$$

Then $F_n = \mathbb{E}[G^n]$. The Fubini numbers are therefore a genuine Stieltjes moment sequence, not merely a combinatorial counting sequence. The same probability model will yield the weighted Fubini polynomials in Section 8.

2.4 Eulerian form and cumulant Bell polynomials

Let $\langle n \rangle_j$ be the Eulerian number counting permutations of n elements with j descents, and put

$$A_n(t) = \sum_{j=0}^{n-1} \langle n \rangle_j t^j \quad (n \geq 1).$$

The classical rational form of the negative polylogarithm is

$$\text{Li}_{-n}(t) = \frac{t A_n(t)}{(1-t)^{n+1}}.$$

Using the symmetry of the Eulerian numbers gives

$$F_n = 2^{n-1} A_n\left(\frac{1}{2}\right) = \sum_{j=0}^{n-1} 2^j \langle n \rangle_j, \quad n \geq 1. \quad (1)$$

The moment generating function of G is $\mathcal{F}(z)$. Its cumulant generating function is

$$K(z) = \log \mathcal{F}(z) = -\log 2 - \log \left(1 - \frac{e^z}{2}\right) = \sum_{m \geq 1} \frac{e^{mz}}{m 2^m} - \log 2.$$

Consequently the r th cumulant is

$$\kappa_r = K^{(r)}(0) = \text{Li}_{1-r}\left(\frac{1}{2}\right), \quad r \geq 1. \quad (2)$$

Moment–cumulant conversion now gives a second Bell-polynomial formula.

Proposition 2.6 (Moment–cumulant formula). *For $n \geq 0$,*

$$F_n = Y_n(\kappa_1, \dots, \kappa_n), \quad \kappa_r = \text{Li}_{1-r}\left(\frac{1}{2}\right).$$

For $r \geq 2$, $\kappa_r = 2F_{r-1}$, while $\kappa_1 = F_0 = 1$.

Proof. The defining identity for complete exponential Bell polynomials is precisely

$$\exp\left(\sum_{r \geq 1} \kappa_r \frac{z^r}{r!}\right) = \sum_{n \geq 0} Y_n(\kappa_1, \dots, \kappa_n) \frac{z^n}{n!}.$$

The left-hand side is $\exp K(z) = \mathcal{F}(z)$, so coefficient comparison proves the first formula. The relation with F_{r-1} follows from Theorem 2.5, noting separately the exceptional 0^0 contribution at $r = 1$. $\square$

3 The exact pole lattice

3.1 Location and residues

The singularities of $\mathcal{F}(z) = (2 - e^z)^{-1}$ are the solutions of $e^z = 2$:

$$\zeta_k = \rho + 2\pi i k, \quad k \in \mathbb{Z}.$$

Since

$$\left. \frac{d}{dz} (2 - e^z) \right|_{z=\zeta_k} = -2,$$

every pole is simple and

$$\operatorname{Res}_{z=\zeta_k} \mathcal{F}(z) = -\frac{1}{2}. \quad (3)$$

The moduli $|\zeta_k|$ increase strictly with $|k|$. Thus the real pole $\zeta_0 = \rho$ is dominant, followed by the conjugate pair $\zeta_{\pm 1}$, then $\zeta_{\pm 2}$, and so forth.

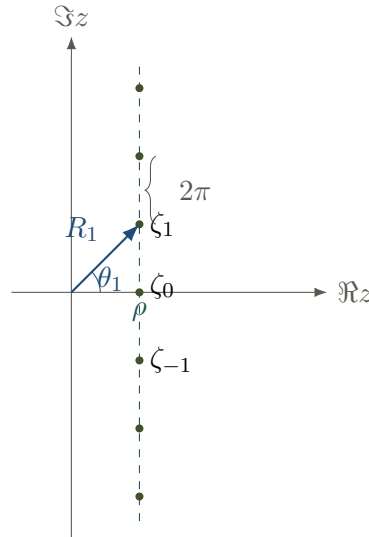


Figure 1: The poles of $(2 - e^z)^{-1}$ lie on the vertical line $\Re z = \rho$. Their equal residues and increasing moduli make the pole ordering identical to the exponential-sector ordering. The drawing rescales the vertical direction for legibility.

3.2 A global Mittag-Leffler decomposition

Put $w = z - \rho$. Since $e^z = 2e^w$,

$$\mathcal{F}(z) = \frac{1}{2(1 - e^w)} = \frac{1}{4} - \frac{1}{4} \coth\left(\frac{w}{2}\right). \quad (4)$$

The paired partial-fraction expansion

$$\coth\left(\frac{w}{2}\right) = \frac{2}{w} + 4w \sum_{k \geq 1} \frac{1}{w^2 + 4\pi^2 k^2} = 2 \sum_{k \in \mathbb{Z}}^{\text{sym}} \frac{1}{w - 2\pi i k} \quad (5)$$

is locally uniformly convergent away from its poles. Combining Equations (4) and (5) yields the exact global decomposition.

Theorem 3.1 (Exact Mittag–Leffler expansion). *For $z \notin \rho + 2\pi i\mathbb{Z}$,*

$$\frac{1}{2 - e^z} = \frac{1}{4} - \frac{1}{2} \sum_{k \in \mathbb{Z}}^{\text{sym}} \frac{1}{z - \zeta_k}. \quad (6)$$

The paired series converges locally uniformly on the complement of the pole lattice.

Proof. Pairing k with $-k$ gives

$$\frac{1}{w - 2\pi i k} + \frac{1}{w + 2\pi i k} = \frac{2w}{w^2 + 4\pi^2 k^2} = O(k^{-2})$$

uniformly on compact w -sets. Hence the paired sum is locally normally convergent away from the poles. The classical product $\sinh(w/2) = (w/2) \prod_{k \geq 1} (1 + w^2/(4\pi^2 k^2))$ and logarithmic differentiation give Equation (5). Substitution into Equation (4) proves Equation (6). $\square$

3.3 Exact coefficient extraction

For a fixed nonzero ζ ,

$$-\frac{1}{2(z - \zeta)} = \frac{1}{2\zeta} \frac{1}{1 - z/\zeta} = \frac{1}{2} \sum_{n \geq 0} \frac{z^n}{\zeta^{n+1}}.$$

When $n \geq 1$, the resulting sum over k is absolutely convergent because its terms are $O(|k|^{-n-1})$.

Theorem 3.2 (Exact all-pole coefficient formula). *For every integer $n \geq 1$,*

$$F_n = \frac{n!}{2} \sum_{k \in \mathbb{Z}} \frac{1}{(\rho + 2\pi i k)^{n+1}}. \quad (7)$$

For $n = 0$, the regularized unified formula is

$$F_0 = \frac{1}{4} + \frac{1}{2} \sum_{k \in \mathbb{Z}}^{\text{sym}} \frac{1}{\rho + 2\pi i k} = 1. \quad (8)$$

Proof. Take the coefficient of z^n in Equation (6) and multiply by $n!$. For $n \geq 1$, absolute convergence justifies termwise coefficient extraction. For $n = 0$, the symmetric sum is

$$\sum_{k \in \mathbb{Z}}^{\text{sym}} \frac{1}{\rho + 2\pi i k} = \frac{1}{2} \coth\left(\frac{\rho}{2}\right).$$

Because $e^\rho = 2$, one has $\coth(\rho/2) = (2 + 1)/(2 - 1) = 3$, so the right-hand side of Equation (8) is $1/4 + 3/4 = 1$. $\square$

Remark 3.3. The separate constant $1/4$ matters only at $n = 0$. This is why the clean bilateral pole formula is usually stated for $n \geq 1$. No regularization is needed at any positive order.

3.4 A second proof by Poisson summation

The geometric-moment formula supplies an independent route to the same identity and simultaneously permits a continuous order parameter.

Theorem 3.4 (Lipschitz–Poisson formula). *For $\nu \in \mathbb{C}$ with $\Re \nu > 0$, define*

$$\mathcal{F}(\nu) := \frac{1}{2} \sum_{m \geq 1} m^\nu e^{-\rho m} = \frac{1}{2} \text{Li}_{-\nu} \left(\frac{1}{2} \right),$$

using the real logarithm in m^ν . Then

$$\mathcal{F}(\nu) = \frac{\Gamma(\nu + 1)}{2} \sum_{k \in \mathbb{Z}} (\rho + 2\pi i k)^{-\nu-1}, \quad (9)$$

where the complex powers use the principal logarithm. At every integer $\nu = n \geq 1$, this is Equation (7).

Proof. Set

$$f_\nu(x) = \begin{cases} x^\nu e^{-\rho x}, & x > 0, \\ 0, & x \leq 0. \end{cases}$$

For $\Re \nu > 0$ the function is continuous at the origin, integrable, and has sufficient decay and local regularity for the classical Poisson summation formula (or for its standard Abel-regularized form followed by a limit). With the Fourier-transform convention $\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} dx$,

$$\widehat{f}_\nu(\xi) = \int_0^\infty x^\nu e^{-(\rho + 2\pi i \xi)x} dx = \frac{\Gamma(\nu + 1)}{(\rho + 2\pi i \xi)^{\nu+1}}.$$

The Fourier series is absolutely convergent because $\Re(\nu + 1) > 1$. Hence

$$\sum_{m \in \mathbb{Z}} f_\nu(m) = \sum_{k \in \mathbb{Z}} \widehat{f}_\nu(k).$$

The left side is $\sum_{m \geq 1} m^\nu e^{-\rho m}$. Division by two proves Equation (9). $\square$

Interpretation

The exact pole transseries is not an accident restricted to integer coefficients. It is the sampling duality between an exponentially damped power on the positive half-line and a vertical reciprocal-power lattice in the Fourier domain. The integer Fubini numbers are special values of this continuous-order identity.

3.5 Hurwitz-zeta and differential forms

Let

$$a := \frac{\rho}{2\pi i} = -\frac{i\rho}{2\pi}.$$

Splitting the bilateral sum into $k \geq 0$ and $k \leq -1$ gives, for $n \geq 1$,

$$F_n = \frac{n!}{2(2\pi i)^{n+1}} [\zeta(n+1, a) + (-1)^{n+1} \zeta(n+1, 1-a)], \quad (10)$$

where the zeta symbols on the right are Hurwitz zeta functions. This form is occasionally convenient for symbolic manipulation, although the conjugate pole-pair form developed below is numerically more transparent.

Define

$$S_s(\rho) := \sum_{k \in \mathbb{Z}}^{\text{sym}} (\rho + 2\pi i k)^{-s}.$$

Then

$$S_1(\rho) = \frac{1}{2} \coth\left(\frac{\rho}{2}\right), \quad S_{s+1}(\rho) = -\frac{1}{s} \frac{d}{d\rho} S_s(\rho).$$

Consequently, for $n \geq 1$,

$$F_n = \frac{(-1)^n}{4} \frac{d^n}{dx^n} \coth\left(\frac{x}{2}\right) \Big|_{x=\rho}. \quad (11)$$

The derivative-polynomial description of $\coth$ is another route from the pole lattice to Eulerian polynomials and hence to Equation (1).

4 The complete exponential transseries

4.1 Complex and real sector coordinates

Define

$$u_k := \frac{\rho}{\zeta_k} = \frac{\rho}{\rho + 2\pi i k}, \quad k \in \mathbb{Z}. \quad (12)$$

Thus $u_0 = 1$, $u_{-k} = \overline{u_k}$, and $|u_k| < 1$ for $k \neq 0$. For $k \geq 1$, write

$$R_k := |\zeta_k| = \sqrt{\rho^2 + 4\pi^2 k^2}, \quad (13)$$

$$q_k := |u_k| = \frac{\rho}{R_k}, \quad (14)$$

$$\alpha_k := -\log q_k = \frac{1}{2} \log \left(1 + \frac{4\pi^2 k^2}{\rho^2} \right), \quad (15)$$

$$\theta_k := \arg \zeta_k = \arctan \left(\frac{2\pi k}{\rho} \right). \quad (16)$$

Then

$$u_k = q_k e^{-i\theta_k} = e^{-\alpha_k - i\theta_k}, \quad u_{-k} = q_k e^{i\theta_k}.$$

The real actions satisfy $0 < \alpha_1 < \alpha_2 < \dots$ and $\alpha_k \rightarrow \infty$.

Theorem 4.1 (Exact pole-lattice transseries). *For every $n \geq 1$, put*

$$T_n := \frac{2\rho^{n+1}}{n!} F_n.$$

Then the following are exact identities:

$$T_n = \sum_{k \in \mathbb{Z}} u_k^{n+1}, \quad (17)$$

$$= 1 + 2 \sum_{k \geq 1} q_k^{n+1} \cos((n+1)\theta_k), \quad (18)$$

$$F_n = \frac{n!}{2\rho^{n+1}} \left[1 + 2 \sum_{k \geq 1} e^{-(n+1)\alpha_k} \cos((n+1)\theta_k) \right]. \quad (19)$$

Both series are absolutely convergent.

Proof. Multiply Equation (7) by $2\rho^{n+1}/n!$ to obtain Equation (17). Pairing k and $-k$ gives

$$u_k^{n+1} + u_{-k}^{n+1} = 2q_k^{n+1} \cos((n+1)\theta_k),$$

which proves the real forms. Absolute convergence follows from $q_k^{n+1} = O(k^{-n-1})$. $\square$

Remark 4.2 (Why this is a transseries). Each $k \neq 0$ contributes a transmonomial $e^{-n\alpha|k|} e^{\mp i n \theta|k|}$ with an additional constant factor u_k . The support is locally finite in real action: below any fixed action bound there are only finitely many k . The conjugate-pair form is real, while the complex form is algebraically more convenient. Unlike a generic formal transseries, this one converges and sums exactly to the normalized integer sequence.

4.2 The first sectors

The first six action parameters are shown in Table 1. Angles are in radians.

Table 1: Pole moduli, sector ratios, real actions, and phases for $\rho = \log 2$.

k	R_k	q_k	α_k	θ_k
1	6.32130292093929	0.109652580999385	2.21043826586918	1.46092281020826
2	12.5854727138613	0.055075180433750	2.89905611015294	1.51569326525480
3	18.8622960281387	0.036747762813494	3.30367793099960	1.53404028826676
4	25.1422977208456	0.027568967174597	3.59106451657788	1.54322386613582
5	31.4235722527350	0.022058191697146	3.81407124054088	1.54873634591964
6	37.7054835106352	0.018383193000679	3.99631845566911	1.55241209822855

Thus the beginning of the exact expansion is

$$\begin{aligned} F_n = \frac{n!}{2\rho^{n+1}} & \left[1 + 2q_1^{n+1} \cos((n+1)\theta_1) + 2q_2^{n+1} \cos((n+1)\theta_2) \right. \\ & \left. + 2q_3^{n+1} \cos((n+1)\theta_3) + \cdots \right]. \end{aligned} \quad (20)$$

The leading correction is not of fixed sign. Since θ_1 is close to $\pi/2$, its sign exhibits a near four-step oscillation with a slowly drifting phase. This is why pointwise error ratios can be much smaller than their envelopes at accidental cancellations.

4.3 Rigorous truncation bounds

For $K \geq 0$, define the K -pair truncation

$$T_{n,K} := 1 + 2 \sum_{k=1}^K q_k^{n+1} \cos((n+1)\theta_k), \quad (21)$$

where an empty sum means $T_{n,0} = 1$.

Theorem 4.3 (Pole-tail bounds). *For $n \geq 1$ and $K \geq 0$,*

$$|T_n - T_{n,K}| \leq B_{n,K} := 2 \sum_{k>K} q_k^{n+1}, \quad (22)$$

$$\leq 2 \left(\frac{\rho}{2\pi} \right)^{n+1} \zeta(n+1, K+1), \quad (23)$$

where $\zeta(s, a)$ is the Hurwitz zeta function. If $K \geq 1$, then also

$$|T_n - T_{n,K}| \leq \frac{2}{n} \left(\frac{\rho}{2\pi} \right)^{n+1} K^{-n}. \quad (24)$$

For the leading approximation $K = 0$ one has the uniform estimate

$$|T_n - 1| \leq \frac{\rho^2}{12} = 0.0400377511598501 \dots \quad (n \geq 1). \quad (25)$$

Proof. The triangle inequality applied to the omitted conjugate pairs gives Equation (22). Since

$$q_k = \frac{\rho}{\sqrt{\rho^2 + 4\pi^2 k^2}} \leq \frac{\rho}{2\pi k},$$

summing the majorant proves Equation (23). For $K \geq 1$ and $s = n + 1$,

$$\sum_{k=K+1}^{\infty} k^{-s} \leq \int_K^{\infty} x^{-s} dx = \frac{K^{-n}}{n},$$

which proves Equation (24). Finally, $\rho/(2\pi) < 1$ and $\zeta(n+1) \leq \zeta(2)$, so

$$B_{n,0} \leq 2 \left(\frac{\rho}{2\pi} \right)^2 \zeta(2) = 2 \frac{\rho^2}{4\pi^2} \frac{\pi^2}{6} = \frac{\rho^2}{12}.$$

□

Corollary 4.4 (Successive asymptotic levels). *As $n \rightarrow \infty$,*

$$F_n \sim \frac{n!}{2\rho^{n+1}}, \quad (26)$$

$$F_n = \frac{n!}{2\rho^{n+1}} [1 + 2q_1^{n+1} \cos((n+1)\theta_1) + O(q_2^{n+1})]. \quad (27)$$

More generally, after K conjugate pairs the relative remainder is bounded by Equation (23) and is therefore $O((\rho/(2\pi(K+1)))^{n+1})$ for fixed K .

Proof. The first statement follows from Equation (25) together with $B_{n,0} \rightarrow 0$. For the second, separate the $k = 1$ pair and apply Equation (23) with $K = 1$. The general case is identical. □

4.4 No hidden local inverse-power coefficients

Near a pole ζ_k ,

$$\mathcal{F}(z) = -\frac{1}{2(z - \zeta_k)} + H_k(z),$$

where H_k is analytic near ζ_k . The singular term contributes exactly $\frac{1}{2}\zeta_k^{-n-1}$ to the ordinary Taylor coefficient. There is no expansion of the residue in n^{-1} . Once the singularity at ζ_k has been subtracted, the analytic remainder is controlled by other singularities, not by an infinite local series at the same pole. This elementary observation explains the constant sector amplitudes in Equation (17). The inverse-power coefficients that appear after elementary normalization will come entirely from $n!$.

5 The factorial fluctuation and its general coefficients

5.1 Exact elementary normalization

Define the scaled gamma function for $z > 0$ by

$$\Gamma^*(z) := \frac{\Gamma(z)}{\sqrt{2\pi} z^{z-1/2} e^{-z}}. \quad (28)$$

At a positive integer n this may be written in the especially useful form

$$n! = \sqrt{2\pi n} \left(\frac{n}{e}\right)^n \Gamma^*(n). \quad (29)$$

Introduce the elementary leading scale

$$\mathcal{A}(n) := \frac{\sqrt{2\pi n}}{2\rho} \left(\frac{n}{e\rho}\right)^n. \quad (30)$$

Combining Equations (17) and (29) gives the exact factorization

$$\boxed{F_n = \mathcal{A}(n)\Gamma^*(n)T_n, \quad T_n = \sum_{k \in \mathbb{Z}} u_k^{n+1}.} \quad (31)$$

The transseries problem has therefore split into two independent pieces: T_n is an exact convergent exponential sum, while $\Gamma^*(n)$ carries the common inverse-power fluctuation.

5.2 Binet's exact Borel–Laplace representation

For $\Re z > 0$, Binet's formula can be written as

$$\log \Gamma^*(z) = \int_0^\infty e^{-zt} h(t) dt, \quad h(t) := \frac{1}{t} \left(\frac{1}{e^t - 1} - \frac{1}{t} + \frac{1}{2} \right). \quad (32)$$

The apparent singularity of h at $t = 0$ is removable. Its Taylor expansion is

$$h(t) = \sum_{m \geq 1} \frac{B_{2m}}{(2m)!} t^{2m-2} = \frac{1}{12} - \frac{t^2}{720} + \frac{t^4}{30240} - \dots \quad (33)$$

The nearest complex singularities of h are at $t = \pm 2\pi i$.

Proposition 5.1 (Exact summation of the logarithmic Stirling series). *For $z > 0$, Equation (32) is an exact positive-axis Borel–Laplace representation. Watson's lemma gives*

$$\log \Gamma^*(z) \sim \sum_{m \geq 1} \frac{B_{2m}}{2m(2m-1)z^{2m-1}} = \sum_{r \geq 1} \frac{b_r}{z^r}, \quad (34)$$

where

$$b_r := \frac{B_{r+1}}{r(r+1)}. \quad (35)$$

Thus $b_{2m} = 0$ and $b_{2m-1} = B_{2m}/[2m(2m-1)]$.

Proof. The expansion Equation (33) follows from the Bernoulli generating function after the singular terms $t^{-1} - 1/2$ are removed. Termwise Laplace integration formally gives

$$\int_0^\infty e^{-zt} \frac{B_{2m}}{(2m)!} t^{2m-2} dt = \frac{B_{2m}(2m-2)!}{(2m)!z^{2m-1}} = \frac{B_{2m}}{2m(2m-1)z^{2m-1}}.$$

Watson's lemma makes this a Poincaré expansion. The exact integral is analytic and unambiguous on the positive Laplace ray because the poles of h lie off that ray. Binet's formula identifies it with $\log \Gamma^*(z)$. $\square$

Remark 5.2. The singularities $\pm 2\pi i, \pm 4\pi i, \dots$ of the Borel kernel control the factorial divergence and hyperasymptotic behavior of Stirling's series. They are conceptually distinct from the poles $\rho + 2\pi i k$ of the Fubini EGF. Formula Equation (31) keeps these two singularity geometries separate.

5.3 Complete Bell-polynomial formula for the Stirling coefficients

Define c_j by the formal exponential

$$C(t) := \exp\left(\sum_{r \geq 1} b_r t^r\right) = \sum_{j \geq 0} c_j t^j. \quad (36)$$

Then

$$\Gamma^*(n) \sim \sum_{j \geq 0} \frac{c_j}{n^j}. \quad (37)$$

The next theorem gives several equivalent all-orders formulae.

Theorem 5.3 (General Stirling coefficient formulae). *Let $c_0 = 1$. For every $j \geq 1$,*

$$c_j = [t^j] \exp\left(\sum_{r \geq 1} \frac{B_{r+1}}{r(r+1)} t^r\right), \quad (38)$$

$$= \frac{1}{j!} Y_j(1!b_1, 2!b_2, \dots, j!b_j), \quad (39)$$

$$= \sum_{\substack{m_1, \dots, m_j \geq 0 \\ m_1 + 2m_2 + \dots + jm_j = j}} \prod_{r=1}^j \frac{b_r^{m_r}}{m_r!}, \quad (40)$$

$$= \sum_{\substack{\nu_1, \nu_2, \dots \geq 0 \\ \sum_{m \geq 1} (2m-1)\nu_m = j}} \prod_{m \geq 1} \frac{1}{\nu_m!} \left(\frac{B_{2m}}{2m(2m-1)}\right)^{\nu_m}. \quad (41)$$

Only finitely many factors occur in Equation (41). The coefficients also satisfy the triangular recurrences

$$c_j = \frac{1}{j} \sum_{r=1}^j \frac{B_{r+1}}{r+1} c_{j-r}, \quad (42)$$

$$c_j = \frac{1}{j} \sum_{m=1}^{\lfloor (j+1)/2 \rfloor} \frac{B_{2m}}{2m} c_{j-2m+1}. \quad (43)$$

In particular every c_j is rational and is effectively computable from $c_0, \dots, c_{j-1}$.

Proof. Formula Equation (38) is the definition Equation (36) with Equation (35). Comparing Equation (36) with the defining generating function of the complete exponential Bell polynomial gives Equation (39). Expanding each factor $\exp(b_r t^r) = \sum_{m_r \geq 0} b_r^{m_r} t^{rm_r} / m_r!$ and collecting the coefficient of t^j gives Equation (40). Since $b_r = 0$ for even r , reindexing odd parts as $r = 2m - 1$ gives Equation (41).

For the recurrence, logarithmically differentiate Equation (36):

$$C'(t) = C(t) \sum_{r \geq 1} r b_r t^{r-1}.$$

The coefficient of t^{j-1} is

$$j c_j = \sum_{r=1}^j r b_r c_{j-r} = \sum_{r=1}^j \frac{B_{r+1}}{r+1} c_{j-r},$$

which proves Equation (42). Deleting the zero even- r terms gives Equation (43). $\square$

Table 2: The first Stirling coefficients c_j in $\Gamma^*(n) \sim \sum c_j n^{-j}$.

j	c_j	j	c_j
0	1	7	$-534703531/902961561600$
1	$1/12$	8	$-4483131259/86684309913600$
2	$1/288$	9	$432261921612371/514904800886784000$
3	$-139/51840$	10	$6232523202521089/86504006548979712000$
4	$-571/2488320$	11	$-25834629665134204969/13494625021640835072000$
5	$163879/209018880$	12	$-1579029138854919086429/9716130015581401251840000$
6	$5246819/75246796800$		

5.4 Divergence versus exact reconstruction

The Bernoulli asymptotic

$$B_{2m} = (-1)^{m-1} \frac{2(2m)!}{(2\pi)^{2m}} \zeta(2m)$$

shows that the logarithmic coefficients grow factorially. Hence neither Equation (34) nor Equation (37) is a convergent power series in $1/n$. This does not impair the exact result:

1. use $n!$ in Equation (19) and no Stirling expansion is present;
2. use $\Gamma^*(n)$ in Equation (31) and the fluctuation is exact;
3. use finitely many c_j for a Poincaré approximation;
4. use the Binet integral Equation (32) to reconstruct the positive-axis Borel sum exactly.

This hierarchy is important when the phrase “full transseries” is used: the complete pole content is convergent, while the standard factorial fluctuation has the familiar asymptotic status of Stirling’s series.

6 The full mixed transseries and rigorous two-parameter errors

6.1 Complex actions and the complete coefficient array

Define the principal complex actions

$$\Lambda_k := \text{Log}\left(\frac{\zeta_k}{\rho}\right) = \begin{cases} 0, & k = 0, \\ \alpha_k + i\theta_k, & k > 0, \\ \alpha_{|k|} - i\theta_{|k|}, & k < 0. \end{cases} \quad (44)$$

Then $u_k = e^{-\Lambda_k}$ and

$$u_k^{n+1} = e^{-n\Lambda_k} e^{-\Lambda_k}.$$

Theorem 6.1 (Full elementary-scale transseries). *As $n \rightarrow +\infty$ through the positive integers,*

$$F_n \sim \mathcal{A}(n) \sum_{k \in \mathbb{Z}} e^{-n\Lambda_k} \sum_{j \geq 0} \frac{C_{j,k}}{n^j}, \quad C_{j,k} = c_j e^{-\Lambda_k} = c_j \frac{\rho}{\rho + 2\pi i k}. \quad (45)$$

Equivalently, in conjugate-pair form,

$$F_n \sim \mathcal{A}(n) \left(\sum_{j \geq 0} \frac{c_j}{n^j} \right) \left[1 + 2 \sum_{k \geq 1} q_k^{n+1} \cos((n+1)\theta_k) \right]. \quad (46)$$

For each fixed pole sector k , every algebraic coefficient is obtained from the principal-sector coefficient by multiplication by ρ/ζ_k .

Proof. Insert the Poincaré expansion $\Gamma^*(n) \sim \sum_{j \geq 0} c_j n^{-j}$ into the exact factorization Equation (31). Next use $T_n = \sum_k e^{-n\Lambda_k} e^{-\Lambda_k}$. Since the pole sum is absolutely convergent for each $n \geq 1$, the coefficient attached to $e^{-n\Lambda_k} n^{-j}$ is exactly $c_j e^{-\Lambda_k}$. Pairing conjugates gives Equation (46). $\square$

Corollary 6.2 (General coefficient formulae in every sector). *For $j \geq 0$ and $k \in \mathbb{Z}$,*

$$C_{j,k} = \frac{\rho}{\zeta_k} \frac{1}{j!} Y_j(1!b_1, \dots, j!b_j), \quad (47)$$

$$= \frac{\rho}{\zeta_k} \sum_{m_1+2m_2+\dots+jm_j=j} \prod_{r=1}^j \frac{b_r^{m_r}}{m_r!}, \quad (48)$$

$$C_{0,k} = \frac{\rho}{\zeta_k}, \quad C_{j,k} = \frac{1}{j} \sum_{r=1}^j \frac{B_{r+1}}{r+1} C_{j-r,k}. \quad (49)$$

Proof. Multiply the corresponding formulae of Theorem 5.3 by ρ/ζ_k . The sector factor is independent of j , so the same triangular recurrence is inherited unchanged. $\square$

6.2 A rigorous mixed truncation theorem

Let

$$S_J(n) := \sum_{j=0}^{J-1} \frac{c_j}{n^j}, \quad J \geq 1, \quad (50)$$

and retain the pole truncation $T_{n,K}$ from Equation (21). Define

$$G_J(n) := \begin{cases} \frac{1}{\pi^2 n}, & J = 1, \\ \frac{(1 + \zeta(J))\Gamma(J)}{(2\pi)^{J+1} n^J}, & J \geq 2. \end{cases} \quad (51)$$

Here $\zeta(J)$ is the Riemann zeta function.

Theorem 6.3 (Two-parameter certified remainder). *For integers $n \geq 1$, $J \geq 1$, and $K \geq 0$,*

$$\left| \frac{F_n}{\mathcal{A}(n)} - S_J(n)T_{n,K} \right| \leq G_J(n)(1 + B_{n,0}) + |S_J(n)|B_{n,K}, \quad (52)$$

where $B_{n,K}$ may be taken from Equations (22) to (24). Consequently, for fixed J, K ,

$$\frac{F_n}{\mathcal{A}(n)} = S_J(n)T_{n,K} + O(n^{-J}) + O(q_{K+1}^{n+1}). \quad (53)$$

Proof. For positive real n , the standard scaled-gamma remainder bound [3, §5.11(ii)] gives

$$|\Gamma^*(n) - S_J(n)| \leq G_J(n). \quad (54)$$

For $J \geq 2$ this is the positive-axis specialization of the classical bound

$$|R_J(z)| \leq \frac{(1 + \zeta(J))\Gamma(J)}{2(2\pi)^{J+1}|z|^J} \left(1 + \min(\sec(\arg z), 2\sqrt{J}) \right);$$

at $z = n > 0$ the last parenthesis is 2. For $J = 1$, the prescribed replacement of $1 + \zeta(J)$ by 4 gives $1/(\pi^2 n)$.

Using the exact identity $F_n/\mathcal{A}(n) = \Gamma^*(n)T_n$,

$$\Gamma^*(n)T_n - S_J(n)T_{n,K} = (\Gamma^*(n) - S_J(n))T_n + S_J(n)(T_n - T_{n,K}).$$

Now $|T_n| \leq 1 + B_{n,0}$ and $|T_n - T_{n,K}| \leq B_{n,K}$. The triangle inequality proves Equation (52); the fixed-truncation asymptotic statement follows. $\square$

Interpretation

The two truncation directions behave differently.

- Increasing K adds terms of a convergent series and monotonically improves the rigorous envelope $B_{n,K}$.
- Increasing J traverses a divergent Poincaré series. It improves the approximation only until the terms approach their least magnitude; the exact Binet integral remains available beyond that point.

Thus pole truncation is ordinary convergence, whereas algebraic truncation is asymptotic optimization.

6.3 Numerical behavior

The next table records the actual absolute relative error

$$\left| \frac{n!}{2\rho^{n+1}} \frac{T_{n,K}}{F_n} - 1 \right| = \left| \frac{T_{n,K}}{T_n} - 1 \right|$$

for several n and numbers K of retained conjugate pole pairs.

Table 3: Actual relative error of exact-factorial pole truncations.

n	$K = 0$	$K = 1$	$K = 2$	$K = 3$
1	4.06845×10^{-2}	1.62606×10^{-2}	9.98550×10^{-3}	7.18241×10^{-3}
2	9.26902×10^{-4}	7.26028×10^{-5}	1.75705×10^{-5}	6.63865×10^{-6}
3	2.85346×10^{-4}	2.37599×10^{-5}	5.80868×10^{-6}	2.20191×10^{-6}
5	2.80687×10^{-6}	5.88313×10^{-8}	6.03717×10^{-9}	1.23133×10^{-9}
10	5.15518×10^{-11}	1.62477×10^{-14}	1.34341×10^{-16}	4.50497×10^{-18}
20	1.02594×10^{-20}	6.65135×10^{-27}	1.03570×10^{-30}	1.95587×10^{-33}
50	1.38135×10^{-49}	4.00015×10^{-65}	1.28072×10^{-73}	5.70913×10^{-80}
100	2.19123×10^{-97}	9.02847×10^{-128}	1.32425×10^{-145}	2.12149×10^{-158}

The nonmonotone-looking comparison between $n = 2$ and $n = 3$ at $K = 0$ is a phase effect, not a failure of the envelope. The bounds in Theorem 4.3 decrease monotonically with n and K even when the actual oscillatory error experiences cancellation.

For the mixed elementary approximation, representative relative errors are shown in Table 4. Here (J, K) means J Stirling terms and K conjugate pole pairs.

Table 4: Actual relative error of selected mixed truncations.

n	(1, 0)	(3, 0)	(6, 0)	(3, 1)	(6, 1)	(10, 1)
5	1.65042×10^{-2}	2.40206×10^{-5}	2.80967×10^{-6}	2.12725×10^{-5}	6.16319×10^{-8}	5.88588×10^{-8}
10	8.29596×10^{-3}	2.67411×10^{-6}	4.08192×10^{-11}	2.67405×10^{-6}	1.07163×10^{-11}	2.76368×10^{-14}
20	4.15765×10^{-3}	3.34956×10^{-7}	6.23885×10^{-13}	3.34956×10^{-7}	6.23885×10^{-13}	2.26443×10^{-18}
50	1.66526×10^{-3}	2.14490×10^{-8}	3.69757×10^{-15}	2.14490×10^{-8}	3.69757×10^{-15}	3.44988×10^{-22}

7 Nonlinear transseries and multi-index coefficient calculus

7.1 A uniformly small exponential correction

Write

$$T_n = 1 + \Delta_n, \quad \Delta_n := \sum_{k \in \mathbb{Z} \setminus \{0\}} u_k^{n+1}. \quad (55)$$

The proof of Equation (25) gives the stronger absolute-majorant statement

$$\sum_{k \neq 0} |u_k|^{n+1} \leq \frac{\rho^2}{12} < 0.041, \quad n \geq 1. \quad (56)$$

Therefore Taylor expansions of analytic functions around $T_n = 1$ are not merely formal: they converge absolutely and uniformly in $n \geq 1$ whenever their disk of analyticity has radius greater than $\rho^2/12$.

Let $I := \mathbb{Z} \setminus \{0\}$. A *finite multi-index* is a family $\mathbf{m} = (m_k)_{k \in I}$ of nonnegative integers with finite support. Put

$$|\mathbf{m}| := \sum_{k \in I} m_k, \quad \mathbf{m}! := \prod_{k \in I} m_k!, \quad u^{\mathbf{m}} := \prod_{k \in I} u_k^{m_k}.$$

The zero multi-index has $|\mathbf{0}| = 0$, $\mathbf{0}! = 1$, and $u^{\mathbf{0}} = 1$.

Theorem 7.1 (Master analytic-transform formula). *Let Φ be analytic in $|z - 1| < R$ for some $R > \rho^2/12$. Then for every $n \geq 1$,*

$$\boxed{\Phi(T_n) = \sum_{\mathbf{m}} \frac{\Phi(|\mathbf{m}|)(1)}{\mathbf{m}!} (u^{\mathbf{m}})^{n+1}.} \quad (57)$$

The sum extends over all finite multi-indices and is absolutely convergent. It is the complete nonlinear exponential transseries of $\Phi(T_n)$.

Proof. Taylor's theorem gives

$$\Phi(1 + \Delta_n) = \sum_{r \geq 0} \frac{\Phi^{(r)}(1)}{r!} \Delta_n^r.$$

By Equation (56), the series converges absolutely. The multinomial expansion in countably many absolutely summable variables is

$$\Delta_n^r = \sum_{\substack{\mathbf{m} \\ |\mathbf{m}|=r}} \frac{r!}{\mathbf{m}!} \prod_{k \in I} (u_k^{n+1})^{m_k} = \sum_{|\mathbf{m}|=r} \frac{r!}{\mathbf{m}!} (u^{\mathbf{m}})^{n+1}.$$

Substitution cancels $r!$ and proves Equation (57). Absolute convergence follows by repeating the same calculation with absolute values and using a slightly smaller circle inside the disk of analyticity. $\square$

7.2 Powers, reciprocals, and logarithms

Taking $\Phi(z) = z^\sigma$, with the branch analytic in $|z - 1| < 1$, gives

$$T_n^\sigma = \sum_{\mathbf{m}} \frac{(\sigma)^{|\mathbf{m}|}}{\mathbf{m}!} (u^{\mathbf{m}})^{n+1}, \quad \sigma \in \mathbb{C}, \quad (58)$$

where $(\sigma)^r = \sigma(\sigma - 1) \cdots (\sigma - r + 1)$. Two important special cases are

$$\frac{1}{T_n} = \sum_{\mathbf{m}} (-1)^{|\mathbf{m}|} \frac{|\mathbf{m}|!}{\mathbf{m}!} (u^{\mathbf{m}})^{n+1}, \quad (59)$$

$$\log T_n = \sum_{\mathbf{m} \neq \mathbf{0}} (-1)^{|\mathbf{m}|-1} \frac{(|\mathbf{m}| - 1)!}{\mathbf{m}!} (u^{\mathbf{m}})^{n+1}. \quad (60)$$

The coefficient in Equation (60) is the connected, or cumulant, version of the multinomial coefficient. Products of distinct pole sectors and repeated powers of a single sector appear automatically.

7.3 Arbitrary powers of the Fubini numbers

For $\sigma \in \mathbb{C}$, define the generalized Stirling coefficients $d_j(\sigma)$ by

$$\exp\left(\sigma \sum_{r \geq 1} b_r t^r\right) = \sum_{j \geq 0} d_j(\sigma) t^j. \quad (61)$$

The same Bell-polynomial argument as before gives

$$d_j(\sigma) = \frac{1}{j!} Y_j(\sigma 1! b_1, \dots, \sigma j! b_j), \quad (62)$$

$$d_0(\sigma) = 1, \quad d_j(\sigma) = \frac{\sigma}{j} \sum_{r=1}^j \frac{B_{r+1}}{r+1} d_{j-r}(\sigma). \quad (63)$$

Thus $\Gamma^*(n)^\sigma \sim \sum_{j \geq 0} d_j(\sigma) n^{-j}$. Combining this with Equation (58) yields the full coefficient calculus for arbitrary powers.

Corollary 7.2 (Full transseries for F_n^σ). *For any fixed $\sigma \in \mathbb{C}$,*

$$F_n^\sigma \sim \mathcal{A}(n)^\sigma \sum_{\mathbf{m}} \frac{(\sigma)^{|\mathbf{m}|}}{\mathbf{m}!} (u^{\mathbf{m}})^{n+1} \sum_{j \geq 0} \frac{d_j(\sigma)}{n^j}. \quad (64)$$

For positive real F_n the principal real power is understood; the right-hand side supplies its analytic continuation in σ .

7.4 The complete logarithmic transseries

Taking the logarithm of Equation (31) gives an exact additive decomposition:

$$\begin{aligned} \log F_n &= (n + \tfrac{1}{2}) \log n - n - (n + 1) \log \rho + \tfrac{1}{2} \log(2\pi) - \log 2 \\ &\quad + \log \Gamma^*(n) + \log T_n. \end{aligned} \quad (65)$$

Inserting the Binet integral and the convergent multi-index sum yields an entirely exact representation.

Theorem 7.3 (Exact and asymptotic logarithmic forms). *For every $n \geq 1$,*

$$\begin{aligned} \log F_n &= (n + \tfrac{1}{2}) \log n - n - (n + 1) \log \rho + \tfrac{1}{2} \log(2\pi) - \log 2 \\ &\quad + \int_0^\infty e^{-nt} h(t) dt + \sum_{\mathbf{m} \neq \mathbf{0}} (-1)^{|\mathbf{m}|-1} \frac{(|\mathbf{m}|-1)!}{\mathbf{m}!} (u^{\mathbf{m}})^{n+1}. \end{aligned} \quad (66)$$

The integral and multi-index sum converge. Its all-orders Poincaré form is

$$\begin{aligned} \log F_n &\sim (n + \tfrac{1}{2}) \log n - n - (n + 1) \log \rho + \tfrac{1}{2} \log(2\pi) - \log 2 \\ &\quad + \sum_{r \geq 1} \frac{B_{r+1}}{r(r+1)n^r} + \sum_{\mathbf{m} \neq \mathbf{0}} (-1)^{|\mathbf{m}|-1} \frac{(|\mathbf{m}|-1)!}{\mathbf{m}!} (u^{\mathbf{m}})^{n+1}. \end{aligned} \quad (67)$$

Proof. Equation (65), Binet's formula Equation (32), and the absolutely convergent identity Equation (60) prove Equation (66). Replacing the integral by its Watson expansion Equation (34) gives Equation (67). $\square$

The first nonlinear exponential terms are

$$\log T_n = \sum_{k \neq 0} u_k^{n+1} - \frac{1}{2} \sum_{k, \ell \neq 0} (u_k u_\ell)^{n+1} + \frac{1}{3} \sum_{k, \ell, m \neq 0} (u_k u_\ell u_m)^{n+1} - \dots$$

The multi-index form is preferable because it collects repeated products and gives their exact multi-nomial coefficients.

7.5 The ratio of consecutive Fubini numbers

The exact factorial factors cancel almost completely:

$$\mathcal{R}_n := \frac{nF_{n-1}}{F_n} = \rho \frac{T_{n-1}}{T_n}, \quad n \geq 1. \quad (68)$$

For an absolutely convergent multi-index expansion of the numerator we take $n \geq 2$. In terms of $y_k = u_k^{n+1}$,

$$T_n = 1 + \sum_{k \neq 0} y_k, \quad T_{n-1} = 1 + \sum_{k \neq 0} u_k^{-1} y_k.$$

Theorem 7.4 (General ratio coefficients). *For $n \geq 2$,*

$$\frac{\mathcal{R}_n}{\rho} = 1 + \sum_{\mathbf{m} \neq \mathbf{0}} r_{\mathbf{m}} (u^{\mathbf{m}})^{n+1}, \quad (69)$$

where, with $M = |\mathbf{m}|$,

$$r_{\mathbf{m}} = (-1)^{M-1} \frac{(M-1)!}{\mathbf{m}!} \left(\sum_{k \neq 0} m_k u_k^{-1} - M \right). \quad (70)$$

The series is absolutely convergent. In particular,

$$\begin{aligned} \frac{\mathcal{R}_n}{\rho} &= 1 + \sum_{k \neq 0} (u_k^{-1} - 1) u_k^{n+1} + O(q_2^n + q_1^{2n}) \\ &= 1 + 2q_1^n [\cos(n\theta_1) - q_1 \cos((n+1)\theta_1)] + O(q_2^n + q_1^{2n}). \end{aligned} \quad (71)$$

Consequently $F_n/F_{n-1} \sim n/\rho$, with exponentially small rather than algebraic relative corrections.

Proof. Multiply

$$\left(1 + \sum_{k \neq 0} u_k^{-1} y_k \right) \left(1 + \sum_{k \neq 0} y_k \right)^{-1}.$$

For a nonzero multi-index $\mathbf{m}$ of total degree M , the contribution from the constant numerator is $(-1)^M M!/\mathbf{m}!$. Choosing the term $u_k^{-1} y_k$ from the numerator leaves multi-index $\mathbf{m} - \mathbf{e}_k$ in the reciprocal and contributes

$$u_k^{-1} (-1)^{M-1} \frac{(M-1)!}{(\mathbf{m} - \mathbf{e}_k)!} = (-1)^{M-1} \frac{(M-1)!}{\mathbf{m}!} m_k u_k^{-1}.$$

Summing over k and combining the constant-numerator term gives Equation (70). Absolute convergence follows from $\sum_{k \neq 0} |u_k|^n < \infty$ for $n \geq 2$ and from $\sum |u_k|^{n+1} < 1$. Keeping only the linear multi-indices and pairing $k = \pm 1$ gives Equation (71); the next envelopes come from the $k = \pm 2$ linear terms and quadratic products of the first pair. $\square$

8 Weighted Fubini polynomials and geometric moments

8.1 Definition and generating functions

For $x > 0$, define the Fubini polynomial

$$F_n(x) := \sum_{j=0}^n j! \left\{ \begin{matrix} n \\ j \end{matrix} \right\} x^j. \quad (72)$$

The same labelled-block argument as in Theorem 2.3 gives

$$\sum_{n \geq 0} F_n(x) \frac{z^n}{n!} = \frac{1}{1 - x(e^z - 1)} = \frac{1}{1 + x - xe^z}. \quad (73)$$

Put

$$q := \frac{x}{1+x}, \quad \rho_x := -\log q = \log\left(1 + \frac{1}{x}\right). \quad (74)$$

Then

$$F_n(x) = (1-q) \sum_{m \geq 0} q^m m^n. \quad (75)$$

Thus $F_n(x)$ is the n th moment of the geometric distribution $\Pr(G_x = m) = (1-q)q^m$. For $n \geq 1$,

$$F_n(x) = (1-q) \operatorname{Li}_{-n}(q). \quad (76)$$

The Eulerian form is

$$F_n(x) = x(1+x)^{n-1} A_n\left(\frac{x}{1+x}\right), \quad n \geq 1. \quad (77)$$

8.2 Exact poles and the weighted transseries

The poles of Equation (73) are

$$\zeta_k(x) = \rho_x + 2\pi i k, \quad k \in \mathbb{Z},$$

and the residue at every pole is $-1/(1+x)$. Repeating the coth decomposition or Poisson summation gives the following master extension.

Theorem 8.1 (Full weighted pole formula). *For $x > 0$ and $n \geq 1$,*

$$F_n(x) = \frac{n!}{1+x} \sum_{k \in \mathbb{Z}} \frac{1}{(\rho_x + 2\pi i k)^{n+1}}. \quad (78)$$

Let

$$u_k(x) := \frac{\rho_x}{\rho_x + 2\pi i k}, \quad T_n(x) := \sum_{k \in \mathbb{Z}} u_k(x)^{n+1}.$$

Then

$$F_n(x) = \mathcal{A}_x(n) \Gamma^*(n) T_n(x), \quad \mathcal{A}_x(n) := \frac{\sqrt{2\pi n}}{(1+x)\rho_x} \left(\frac{n}{e\rho_x}\right)^n. \quad (79)$$

Its full elementary-scale coefficients are

$$C_{j,k}(x) = c_j \frac{\rho_x}{\rho_x + 2\pi i k}. \quad (80)$$

In particular,

$$F_n(x) \sim \frac{n!}{(1+x)\rho_x^{n+1}} \quad (n \rightarrow \infty, \text{ fixed } x > 0). \quad (81)$$

Proof. Factor the EGF as

$$\frac{1}{1+x-xe^z} = \frac{1}{1+x} \frac{1}{1-qe^z}.$$

With $w = z - \rho_x$, one has $qe^z = e^w$, hence

$$\frac{1}{1+x-xe^z} = \frac{1}{2(1+x)} - \frac{1}{2(1+x)} \coth\left(\frac{w}{2}\right).$$

The coefficient extraction of Theorems 3.1 and 3.2 now applies with residue $-1/(1+x)$ and ρ replaced by ρ_x , proving Equation (78). The remaining statements follow from Equation (29) exactly as before. $\square$

Corollary 8.2 (Uniformity on compact weight intervals). *Fix $0 < a \leq x \leq b < \infty$. For every fixed K and J , the weighted analogues of Equation (53) hold uniformly in $x \in [a, b]$.*

Proof. On $[a, b]$, ρ_x stays in a compact subinterval of $(0, \infty)$. Hence $|u_k(x)|$ is uniformly bounded away from one for every fixed $k \neq 0$, and the tails are uniformly dominated by a constant multiple of $\sum_{k>K} k^{-n-1}$. The scaled-gamma remainder is independent of x . $\square$

Remark 8.3 (Nonuniform endpoint regimes). The fixed- x transseries is not uniform as $x \downarrow 0$. In that limit $\rho_x \rightarrow \infty$ and $|u_1(x)| \rightarrow 1$, so many pole sectors become comparable before the eventual $n \rightarrow \infty$ hierarchy separates them. By contrast, as $x \rightarrow \infty$, $\rho_x \sim x^{-1}$ and the nonreal sectors are extremely suppressed. A double-scaling theory in which n and x vary together is therefore a separate asymptotic problem rather than a formal substitution into the fixed- x result.

8.3 Weighted cumulants and Bell-polynomial reconstruction

The cumulant generating function of the geometric law in Equation (75) is

$$K_x(z) = \log(1 - q) - \log(1 - qe^z).$$

Thus

$$\kappa_r(x) = \text{Li}_{1-r}(q), \quad q = \frac{x}{1+x}, \quad (82)$$

and

$$F_n(x) = Y_n(\kappa_1(x), \dots, \kappa_n(x)). \quad (83)$$

This identity supplies an exact coefficient bridge between ordered set partitions, geometric moments, and the Bell-polynomial calculus used for the Stirling fluctuation. The two appearances of Bell polynomials play different roles: Equation (83) converts probabilistic cumulants into exact moments, whereas Equation (39) exponentiates the asymptotic logarithm of a gamma factor.

9 A general meromorphic-EGF principle and higher-order poles

9.1 Coefficient contributions of arbitrary principal parts

The Fubini case is a particularly transparent instance of a general theorem.

Theorem 9.1 (Meromorphic exponential generating functions). *Let*

$$A(z) = \sum_{n \geq 0} a_n \frac{z^n}{n!}$$

be meromorphic in a neighborhood of the closed disk $|z| \leq R$, with no pole on $|z| = R$. Let its poles in the disk be $\xi_1, \dots, \xi_M$, and write the principal part at ξ_ℓ as

$$\sum_{s=1}^{d_\ell} a_{\ell,-s} (z - \xi_\ell)^{-s}.$$

Then

$$a_n = n! \sum_{\ell=1}^M \sum_{s=1}^{d_\ell} (-1)^s a_{\ell,-s} \binom{n+s-1}{s-1} \xi_\ell^{-n-s} + E_n, \quad (84)$$

where, after subtracting all principal parts, the analytic remainder H obeys

$$|E_n| \leq n! R^{-n} \max_{|z|=R} |H(z)|. \quad (85)$$

In particular, a simple pole of residue r_ℓ contributes exactly

$$-n!r_\ell\xi_\ell^{-n-1},$$

whereas a pole of order d contributes an exponential factor ξ^{-n} multiplied by a polynomial in n of degree at most $d - 1$.

Proof. Subtract the principal parts and call the result $H(z)$. It is analytic on a neighborhood of the closed disk. For a single principal-part term,

$$(z - \xi)^{-s} = (-\xi)^{-s}(1 - z/\xi)^{-s} = (-1)^s \sum_{n \geq 0} \binom{n + s - 1}{s - 1} \xi^{-n-s} z^n.$$

Summing these coefficient contributions and multiplying by $n!$ gives the main term in Equation (84). Cauchy's coefficient estimate on $|z| = R$ gives Equation (85). The binomial coefficient is a polynomial in n of degree $s - 1$, proving the final statement. $\square$

Remark 9.2. For a function with infinitely many poles one applies Theorem 9.1 on expanding disks. If a global Mittag-Leffler decomposition is available, as in Theorem 3.1, the limiting sum may be exact. The theorem also explains the sector ordering: poles are ranked by modulus, while equal-modulus conjugates must be retained together for a real approximation.

9.2 Higher-order Fubini numbers

For an integer $r \geq 1$, define the order- r Fubini numbers by

$$\frac{1}{(2 - e^z)^r} = \sum_{n \geq 0} F_n^{(r)} \frac{z^n}{n!}. \quad (86)$$

Every pole now has order r . Introduce the generalized Bernoulli, or Nørlund, numbers by

$$\left(\frac{t}{e^t - 1} \right)^r = \sum_{j \geq 0} B_j^{(r)} \frac{t^j}{j!}. \quad (87)$$

For example,

$$B_0^{(r)} = 1, \quad B_1^{(r)} = -\frac{r}{2}, \quad B_2^{(r)} = \frac{r(3r - 1)}{12}.$$

Theorem 9.3 (Exact pole transseries for higher order). *For integers $r \geq 1$ and $n \geq 1$,*

$$F_n^{(r)} = \frac{n!}{2^r} \sum_{k \in \mathbb{Z}} \sum_{j=0}^{r-1} \frac{(-1)^j B_j^{(r)}}{j!} \binom{n + r - j - 1}{r - j - 1} \zeta_k^{-n-r+j}. \quad (88)$$

The sums are absolutely convergent. In particular,

$$F_n^{(r)} \sim \frac{n!}{2^r} \binom{n + r - 1}{r - 1} \rho^{-n-r}. \quad (89)$$

Every pole sector has a polynomial amplitude of degree $r - 1$, whose complete coefficients are the generalized Bernoulli numbers in Equation (88).

Proof. Near $z = \zeta_k$, put $w = z - \zeta_k$. Since $e^{\zeta_k} = 2$,

$$\begin{aligned} \frac{1}{(2 - e^z)^r} &= \frac{1}{2^r(1 - e^w)^r} \\ &= \frac{(-1)^r}{2^r} w^{-r} \left(\frac{w}{e^w - 1} \right)^r \\ &= \frac{(-1)^r}{2^r} \sum_{j \geq 0} \frac{B_j^{(r)}}{j!} w^{j-r}. \end{aligned}$$

The principal part consists of $0 \leq j \leq r - 1$. Apply Theorem 9.1 to each principal-part term. If $s = r - j$, then the sign from coefficient extraction is $(-1)^s$; combined with $(-1)^r$ it is $(-1)^j$. This gives exactly the summand in Equation (88).

For completeness, exactness rather than merely an asymptotic sum can be seen without any global convergence subtlety. For $a > 1$ and $n \geq 1$,

$$n![z^n] \frac{1}{a - e^z} = \frac{n!}{a} \sum_{k \in \mathbb{Z}} (\log a + 2\pi i k)^{-n-1}.$$

Moreover,

$$(a - e^z)^{-r} = \frac{(-1)^{r-1}}{(r-1)!} \frac{\partial^{r-1}}{\partial a^{r-1}} (a - e^z)^{-1}.$$

The bilateral series and its first $r - 1$ parameter derivatives converge locally uniformly for $n \geq 1$, so differentiation followed by $a = 2$ proves that the sum of all pole contributions is exact. Finally the unique dominant term is $k = 0, j = 0$, which gives Equation (89). $\square$

Example 9.4 (Order two). Since $B_0^{(2)} = 1$ and $B_1^{(2)} = -1$,

$$F_n^{(2)} = \frac{n!}{4} \sum_{k \in \mathbb{Z}} [(n+1)\zeta_k^{-n-2} + \zeta_k^{-n-1}]. \quad (90)$$

The first values are 1, 2, 8, 44, 308, 2612, 25988, ... The linear polynomial $n + 1$ is the characteristic signature of a double pole.

9.3 What the general theorem says about “full” expansions

The preceding results separate three sources of coefficients that are often mixed together:

1. the *principal-part coefficients* at a generating-function pole, which determine the polynomial sector amplitude;
2. the *pole location*, which determines the exponential action and phase;
3. any *external normalization*, such as replacing $n!$ by an elementary Stirling scale, which supplies a common algebraic series.

For ordinary Fubini numbers the first item is only the residue $-1/2$, so the native pole sectors have no internal $1/n$ series. Formula Equation (45) contains inverse powers only because of the third item. Formula Equation (88) shows explicitly how the answer changes when the first item becomes nontrivial.

10 Derived asymptotic quantities

10.1 Root asymptotics

Dividing Equation (66) by n gives a convenient form for the root growth. In particular,

$$\begin{aligned} F_n^{1/n} = \frac{n}{e\rho} \exp & \left[\frac{\frac{1}{2} \log(2\pi n) - \log(2\rho)}{n} + \frac{1}{12n^2} - \frac{1}{360n^4} + \frac{1}{1260n^6} + O(n^{-8}) \right] \\ & \times \exp \left[\frac{2q_1^{n+1} \cos((n+1)\theta_1) + O(q_2^{n+1} + q_1^{2n+2})}{n} \right]. \end{aligned} \quad (91)$$

The first line is the algebraic expansion; the second is the exponentially small oscillatory correction. At leading order,

$$F_n^{1/n} \sim \frac{n}{e \log 2}.$$

10.2 Successive quotients

The ratio theorem gives

$$\frac{F_n}{F_{n-1}} = \frac{n}{\rho} \{1 - 2q_1^n [\cos(n\theta_1) - q_1 \cos((n+1)\theta_1)] + O(q_2^n + q_1^{2n})\}. \quad (92)$$

There are no algebraic $1/n$ corrections. This cancellation is exact: $n!/(n-1)! = n$, and the common power of ρ contributes $1/\rho$. Only the exponentially small pole corrections survive.

10.3 Logarithmic convexity at large order

For $n \geq 1$, the geometric-moment representation implies log-convexity directly by Cauchy–Schwarz:

$$F_n^2 = \mathbb{E}[G^n]^2 \leq \mathbb{E}[G^{n-1}] \mathbb{E}[G^{n+1}] = F_{n-1} F_{n+1}.$$

The transseries refines the asymptotic size of the log-convexity quotient. Set

$$Q_n := \frac{F_{n-1} F_{n+1}}{F_n^2}.$$

The factorial and dominant-pole factors give $Q_n = (n+1)T_{n-1}T_{n+1}/(nT_n^2)$, hence

$$Q_n = 1 + \frac{1}{n} + O(q_1^n). \quad (93)$$

A complete multi-index expansion follows from Theorem 7.1 after adjoining the shifted sector variables u_k^n and u_k^{n+2} .

11 Computation, verification, and implementation

11.1 Which formula should be used?

The available formulae serve different computational regimes.

1. For small exact n , use the Stirling sum Theorem 2.2 or the binomial recurrence Theorem 2.4; both use exact integers.
2. For high-precision asymptotics with moderate or large n , evaluate a few conjugate pole pairs in polar form and keep $n!$ through $\log \Gamma(n+1)$.
3. For an elementary closed approximation, use $S_J(n)T_{n,K}$ and the certified error Equation (52).

4. For extremely high algebraic accuracy, use the Binet integral rather than forcing the divergent Stirling series past its least term.

The pole-pair form avoids cancellation between huge complex quantities:

$$\log\left(\frac{n!}{2\rho^{n+1}}\right) = \log \Gamma(n+1) - \log 2 - (n+1) \log \rho,$$

followed by multiplication by the real quantity $T_{n,K}$.

11.2 A priori choice of the number of pole pairs

Let

$$\eta := 1 - \frac{\rho^2}{12} = 0.959962248840149 \dots$$

By Equation (25), $|T_n| \geq \eta$. Hence the relative error of the exact-factorial K -pair approximation satisfies

$$\left| \frac{T_{n,K}}{T_n} - 1 \right| \leq \frac{B_{n,K}}{\eta}.$$

For a target relative tolerance ε , it is therefore enough to choose an integer $K \geq 1$ such that

$$\frac{2}{n} \left(\frac{\rho}{2\pi} \right)^{n+1} K^{-n} \leq \eta \varepsilon. \quad (94)$$

Equivalently, one may take

$$K \geq \left[\frac{2}{n\eta\varepsilon} \left(\frac{\rho}{2\pi} \right)^{n+1} \right]^{1/n} \quad (95)$$

and round the right-hand side upward. For an absolute tolerance in the normalized factor T_n , simply omit η . These bounds are intentionally simple and conservative; the exact majorant $B_{n,K} = 2 \sum_{k>K} q_k^{n+1}$ or the Hurwitz-zeta bound is usually sharper.

11.3 Wolfram Language reference implementation

The following definitions compute exact Fubini numbers, pole approximations, Stirling coefficients, and certified pole-tail envelopes.

Listing 1: Reference implementation in Wolfram Language.

```
ClearAll[fubiniExact, fubiniPole, stirlingCoefficient, poleTailBound];

fubiniExact[n_Integer?NonNegative] :=
  Sum[k! StirlingS2[n, k], {k, 0, n}];

fubiniPole[n_Integer?Positive, K_Integer?NonNegative,
  wp_Integer : 80] := Module[{rho = N[Log[2], wp]},
  N[n!/2 Sum[(rho + 2 Pi I k)^(-n - 1), {k, -K, K}], wp]
];

stirlingCoefficient[0] = 1;
stirlingCoefficient[j_Integer?Positive] :=
  stirlingCoefficient[j] = Together[
    1/j Sum[
      BernoulliB[r + 1]/(r + 1) stirlingCoefficient[j - r],
      {r, 1, j}
    ]
];
```

```
poleTailBound[n_Integer?Positive, K_Integer?NonNegative] :=
  2 (Log[2]/(2 Pi))^(n + 1) HurwitzZeta[n + 1, K + 1];

Table[fubiniExact[n], {n, 0, 10}]
Table[stirlingCoefficient[j], {j, 0, 12}]
```

For a real-valued and phase-stable pole calculation one may replace the bilateral complex sum with

```
fubiniPoleReal[n_Integer?Positive, K_Integer?NonNegative,
  wp_Integer : 80] := Module[{rho, term},
  rho = N[Log[2], wp];
  term[k_] := Module[{z = rho + 2 Pi I k},
    (rho/Abs[z])^(n + 1) Cos[(n + 1) Arg[z]]
  ];
  N[n!/(2 rho^(n + 1)) (1 + 2 Sum[term[k], {k, 1, K}]), wp]
];
```

11.4 Internal consistency checks

Several identities provide independent tests of an implementation:

$$\begin{aligned}
 F_n &= \sum_{j=0}^n j! \left\{ \begin{matrix} n \\ j \end{matrix} \right\}, \\
 &= \sum_{j=0}^{n-1} \binom{n}{j} F_j, \\
 &= \frac{1}{2} \sum_{m \geq 0} \frac{m^n}{2^m}, \\
 &= \frac{n!}{2} \sum_{k \in \mathbb{Z}} \zeta_k^{-n-1}, \\
 &= 2^{n-1} A_n(1/2) \quad (n \geq 1).
 \end{aligned}$$

The first two are finite exact integer computations, the third and fourth are independent rapidly convergent numerical sums, and the fifth uses a different combinatorial triangle. Agreement among all five is a strong guard against normalization, indexing, and $n = 0$ errors.

Table 5: Exact initial values used in the numerical checks.

n	F_n	n	F_n
0	1	6	4683
1	1	7	47293
2	3	8	545835
3	13	9	7087261
4	75	10	102247563
5	541	11	1622632573

12 Literature context and relation to the linked research drafts

The numbers studied here have circulated under several names. Gross’s “preferential arrangements” terminology and Good’s study of rankings with ties are standard early modern references [6, 5];

Comtet popularized the name Fubini numbers in the coefficient-calculus literature [2]. Good's Theorem 4 already records the exact reciprocal-pole/cosine expansion underlying Theorems 3.2 and 4.1; the present organization recasts that identity as a convergent pole-lattice transseries and develops its all-orders coefficient, remainder, and nonlinear-transform calculus. Their combinatorial and analytic identities are also catalogued as OEIS A000670 [9]. The appearance of Fubini numbers as coefficients in asymptotic expansions of hypercube resistance was developed by Pippenger [12]; barred preferential arrangements give one of many related higher-order families [1].

The leading-pole estimate is a direct instance of meromorphic singularity analysis, for which [4] provides a broad systematic framework. What is emphasized in the present article is the *complete* pole lattice and its exact Mittag-Leffler sum, together with the resulting transseries algebra. The Stirling fluctuation and its bounds are standard gamma-function asymptotics; useful general sources include [3, 10, 15, 11]. The distinction between a divergent Poincaré series and its summation is classical [7].

The notation and organization deliberately align with the two linked research areas in the ProveIt repository. The first develops Stirling, Bell-polynomial, Faà di Bruno, and inversion coefficient calculus [13]; the second organizes formal series and transseries by their asymptotic scales and algebraic operations [14]. The present paper supplies a worked synthesis: the EGF pole lattice provides the transmonomials, complete Bell polynomials provide all factorial coefficients, and multi-index multinomial calculus provides nonlinear sector products. No claim is made here that the analytic pole and remainder theorems have been machine-checked in Lean.

13 Conclusions and further directions

The Fubini numbers admit a stronger description than the customary leading asymptotic formula suggests. In exact factorial normalization,

$$\frac{2(\log 2)^{n+1}}{n!} F_n = \sum_{k \in \mathbb{Z}} \left(\frac{\log 2}{\log 2 + 2\pi i k} \right)^{n+1},$$

so the complete exponentially small structure is an absolutely convergent sum indexed by every EGF pole. The conjugate-pair form makes each real action and oscillatory phase explicit. Because the poles are simple, their native amplitudes are constants.

Replacing $n!$ by its elementary Stirling scale introduces a common inverse-power series. Its coefficients are

$$c_j = \frac{1}{j!} Y_j(1!b_1, \dots, j!b_j), \quad b_r = \frac{B_{r+1}}{r(r+1)},$$

and the complete mixed coefficient array is the rank-one factorization

$$C_{j,k} = c_j \frac{\log 2}{\log 2 + 2\pi i k}.$$

This factorization is the central all-orders result: the j -dependence is universal across sectors, and the k -dependence is a single pole multiplier. The two-parameter theorem Theorem 6.3 turns any finite rectangle in this coefficient array into a certified approximation.

The same framework remains exact under analytic nonlinear operations on the normalized pole sum, yielding closed multi-index coefficients for powers, reciprocals, logarithms, and ratios. It extends uniformly to positive weights on compact intervals and, after replacing simple poles by multiple poles, to higher-order Fubini families with generalized-Bernoulli polynomial amplitudes.

Several directions are natural continuations.

1. **Uniform small-weight scaling.** For $x \downarrow 0$, $\rho_x \rightarrow \infty$ and the fixed- x sector hierarchy loses uniformity. Regimes such as n/ρ_x fixed should lead to a different continuum approximation to the pole lattice.

2. **Complex order and Stokes geometry.** The continuous-order formula Equation (9) defines sectors for complex ν . Their dominance ordering changes with $\arg \nu$, giving a natural Stokes and anti-Stokes diagram even though the positive real sequence has a globally convergent pole sum.
3. **Hyperasymptotics of the common gamma factor.** The Borel singularities at $\pm 2\pi im$ can be resolved with terminant functions. Combining that hyperasymptotic layer with the exact Fubini pole lattice would produce a two-geometry resurgent description.
4. **Higher-order and barred arrangements.** Formula Equation (88) suggests a general catalogue in which combinatorial constructions are classified by the orders and residues of their meromorphic EGF poles.
5. **Formal verification.** The Stirling-transform EGF, Bell coefficient recurrence, finite pole truncations, and algebraic multi-index identities are plausible first targets. The analytic Mittag-Leffler, Poisson, and remainder layers require a developed complex-analysis interface.

A Explicit derivation of the first algebraic coefficients

The logarithmic coefficients begin

$$b_1 = \frac{1}{12}, \quad b_2 = 0, \quad b_3 = -\frac{1}{360}, \quad b_4 = 0, \quad b_5 = \frac{1}{1260}, \dots$$

Expanding $\exp(\sum b_r t^r)$ by partitions of j gives

$$\begin{aligned} c_0 &= 1, \\ c_1 &= b_1 = \frac{1}{12}, \\ c_2 &= b_2 + \frac{b_1^2}{2!} = \frac{1}{288}, \\ c_3 &= b_3 + b_1 b_2 + \frac{b_1^3}{3!} = -\frac{139}{51840}, \\ c_4 &= b_4 + b_1 b_3 + \frac{b_2^2}{2!} + \frac{b_1^2 b_2}{2!} + \frac{b_1^4}{4!} = -\frac{571}{2488320}, \\ c_5 &= b_5 + b_2 b_3 + b_1 b_4 + \frac{b_1^2 b_3}{2!} + \frac{b_1 b_2^2}{2!} + \frac{b_1^3 b_2}{3!} + \frac{b_1^5}{5!} = \frac{163879}{209018880}. \end{aligned}$$

These examples illustrate why the Bell-polynomial notation is not cosmetic: at order j , one must sum over every partition of j using the admissible odd part sizes. Formula Equation (41) performs that bookkeeping at arbitrary order.

For a fixed pole k , multiplication by ρ/ζ_k gives immediately

$$C_{0,k} = \frac{\rho}{\zeta_k}, \quad C_{1,k} = \frac{\rho}{12\zeta_k}, \quad C_{2,k} = \frac{\rho}{288\zeta_k}, \quad C_{3,k} = -\frac{139\rho}{51840\zeta_k}, \dots$$

No second recurrence in k is needed.

B A proof-oriented transseries dictionary

Table 6: Objects in the Fubini calculation and their analytic roles.

Object	Role
$F_n = \sum_j j! \binom{n}{j}$ $(2 - e^z)^{-1}$	Exact ordered-partition count; a Stirling transform of factorials. Exponential generating function; its meromorphic geometry determines the exponential sectors.
$\zeta_k = \rho + 2\pi i k$	Complete pole lattice. Modulus determines suppression; argument determines oscillation.
$u_k = \rho / \zeta_k$	Dimensionless sector multiplier. The normalized exact transmonomial is u_k^{n+1} .
$\Lambda_k = \text{Log}(\zeta_k / \rho)$	Complex action; $u_k^{n+1} = e^{-n\Lambda_k} e^{-\Lambda_k}$.
$T_n = \sum_k u_k^{n+1}$	Exact, absolutely convergent pole-lattice transseries in factorial normalization.
$\Gamma^*(n)$	Exact common factorial fluctuation in elementary normalization.
$b_r = B_{r+1} / [r(r+1)]$	Coefficients of the logarithmic Stirling series.
c_j	Coefficients after exponentiation; complete Bell polynomials in the b_r .
$C_{j,k} = c_j \rho / \zeta_k$	Complete mixed coefficient array for algebraic order j and pole sector k .
$B_{n,K}$	Rigorous envelope of all omitted pole pairs beyond K .
$G_J(n)$	Rigorous scaled-gamma remainder after J algebraic terms.
finite multi-index $\mathbf{m}$	Records products of nonprincipal pole sectors created by nonlinear operations.

C Continuous-order asymptotics

For real $\nu > 0$, the exact formula Equation (9) may be written

$$\mathcal{F}(\nu) = \frac{\sqrt{2\pi\nu}}{2\rho} \left(\frac{\nu}{e\rho} \right)^\nu \Gamma^*(\nu) \sum_{k \in \mathbb{Z}} u_k^{\nu+1}. \quad (96)$$

Here powers use the principal logarithm. Since $u_{-k} = \overline{u_k}$, the sum is real for real ν . Therefore

$$\begin{aligned} \mathcal{F}(\nu) &= \frac{\sqrt{2\pi\nu}}{2\rho} \left(\frac{\nu}{e\rho} \right)^\nu \Gamma^*(\nu) \\ &\quad \times \left[1 + 2 \sum_{k \geq 1} q_k^{\nu+1} \cos((\nu+1)\theta_k) \right]. \end{aligned} \quad (97)$$

The same coefficients c_j and $C_{j,k}$ apply with n replaced by ν . For complex ν with $\Re \nu > 0$, the bilateral sum remains absolutely convergent, but conjugate pairing no longer produces a real expression and the relative dominance of k and $-k$ depends on $\Im \nu$.

D Sharper elementary pole-tail estimates

The Hurwitz-zeta bound in Equation (23) is simple and uniform. Several refinements are occasionally useful. Since q_k decreases with k ,

$$B_{n,K} = 2q_{K+1}^{n+1} \left[1 + \sum_{r \geq 1} \left(\frac{R_{K+1}}{R_{K+1+r}} \right)^{n+1} \right]. \quad (98)$$

Thus the first omitted conjugate pair gives the correct exponential scale. Using $R_k \geq 2\pi k$ only in the remaining tail yields

$$B_{n,K} \leq 2q_{K+1}^{n+1} + 2 \left(\frac{\rho}{2\pi} \right)^{n+1} \zeta(n+1, K+2). \quad (99)$$

This is often considerably sharper than applying the elementary majorant to the first omitted pair as well.

For $K = 0$, a phase-blind two-sided estimate follows from Equation (25):

$$1 - \frac{\rho^2}{12} \leq \frac{2\rho^{n+1}F_n}{n!} \leq 1 + \frac{\rho^2}{12}, \quad n \geq 1. \quad (100)$$

The lower bound is positive, so logarithms, reciprocals, and arbitrary real powers of the normalized sequence are globally well-defined for all positive integer n .

E Generalized Bernoulli coefficients for multiple poles

The generalized Bernoulli numbers in Equation (87) themselves admit Bell-polynomial formulae. Write

$$\log\left(\frac{t}{e^t - 1}\right) = -\frac{t}{2} - \sum_{m \geq 1} \frac{B_{2m}}{2m(2m)!} t^{2m}.$$

If ℓ_j denotes the coefficient of t^j in this logarithm, then

$$B_j^{(r)} = Y_j(1!r\ell_1, 2!r\ell_2, \dots, j!r\ell_j).$$

Writing $\log(t/(e^t - 1)) = \sum_{m \geq 1} \ell_m t^m$, logarithmic differentiation of Equation (87) gives the triangular recurrence

$$B_0^{(r)} = 1, \quad B_j^{(r)} = r \sum_{m=1}^j \binom{j-1}{m-1} (m!\ell_m) B_{j-m}^{(r)}. \quad (101)$$

Here $\ell_1 = -1/2$, $\ell_{2m} = -B_{2m}/[2m(2m)!]$, and $\ell_{2m+1} = 0$ for $m \geq 1$. Any preferred standard recurrence for Nørlund numbers may be substituted in Equation (88); the pole formula itself requires only the generating definition Equation (87).

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